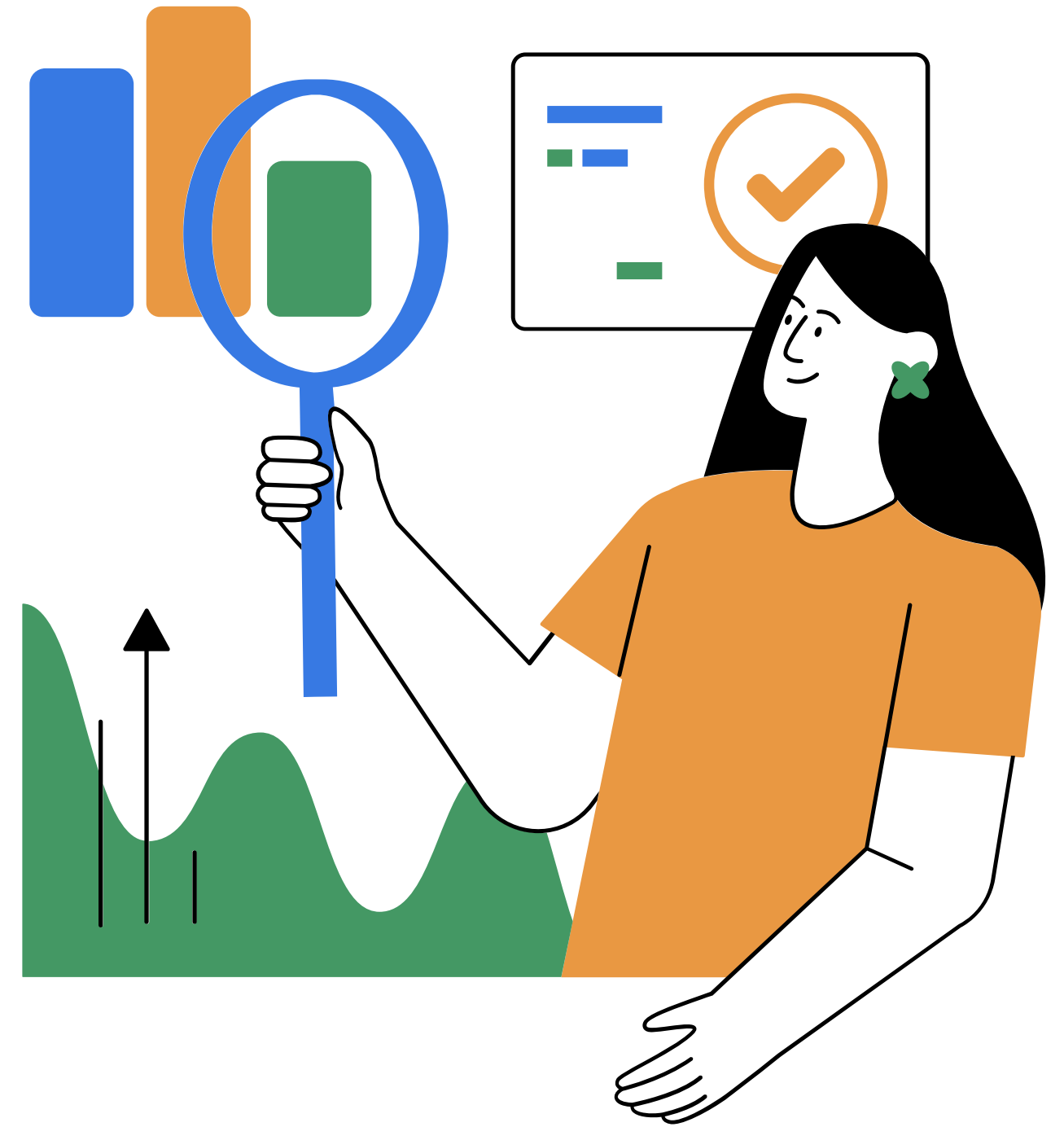
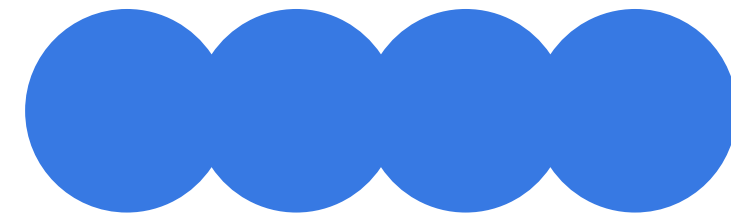


Understanding Cart Abandonment in Digital Retail





Choosing an Industry



Industries we've considered so far



1.

DIGITAL RETAIL

Broader industry with richer data, clearer customer behavior patterns, and more business applications.



2.

TOURISM

Strongly affected by seasonality, location, weather, and external events, making the analysis more complex.



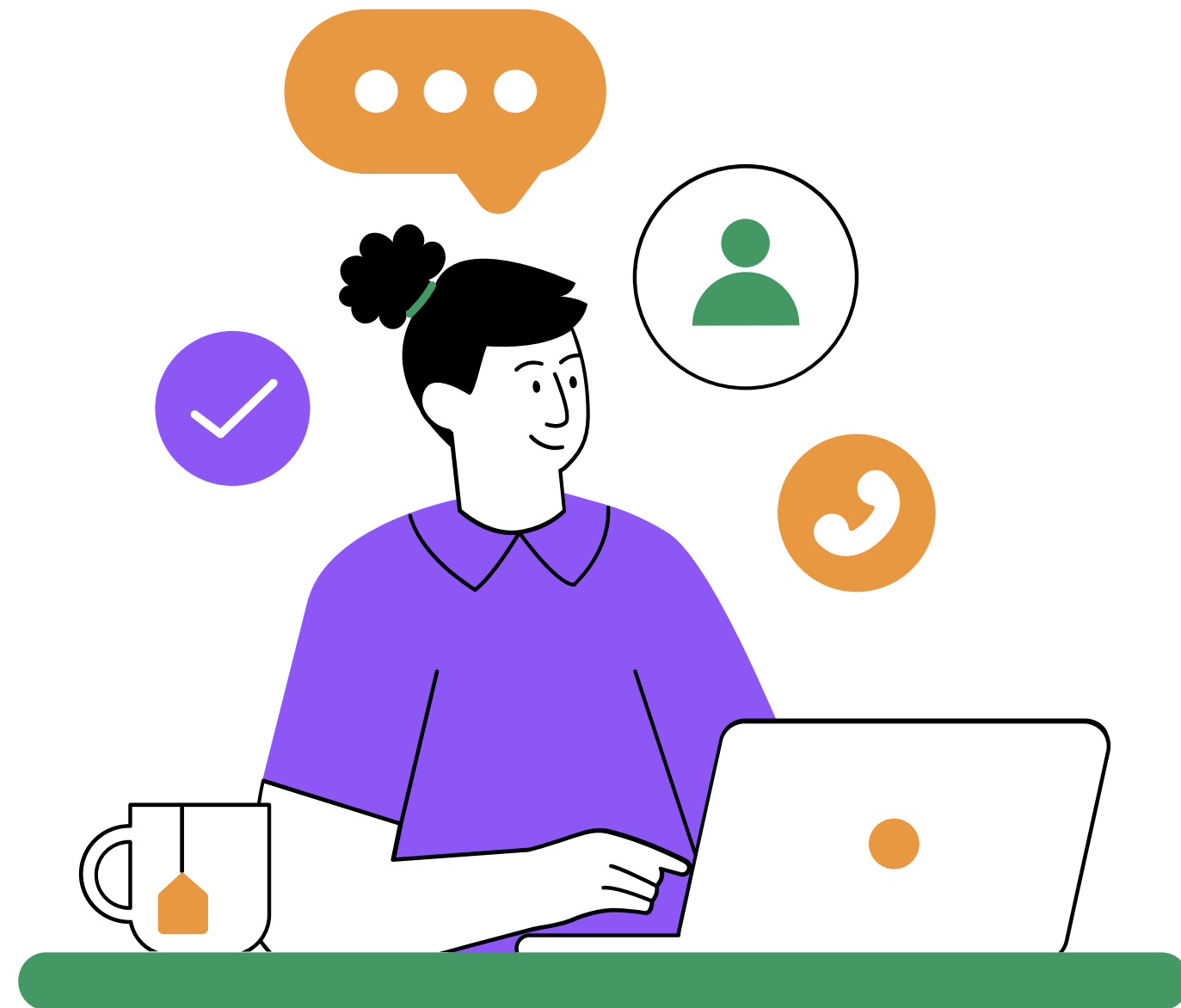
3.

SMARTPHONES

Market trends change quickly due to new releases, brand loyalty, and technology cycles.



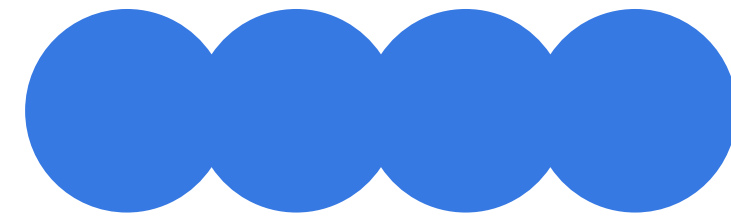
Final Decision



Digital Retail (e-commerce)

Digital retail provides rich customer behavior data, including browsing patterns, product views, cart activity, pricing, and purchase decisions.

It has strong business value because insights can improve conversion rates, reduce lost sales, and support better marketing and product strategies.



Problem

KPIs and Challenges Facing the Digital Retail Industry

Pre-purchasing Challenges

1

Low conversion rates/Cart abandonment :

Many customers browse products but leave without completing a purchase.

2

Difficulty understanding customer intent:

Retailers may struggle to know whether a customer is seriously interested or only browsing.

3

Price sensitivity:

Customers may leave if they find a cheaper or better option elsewhere.

4

Changing customer behavior:

Customer preferences shift quickly based on trends, price, convenience, and product availability.



Business / Operation Challenges

5

Inventory imbalance:

Some products sell out quickly while others remain unsold, making demand planning difficult.

Post-purchase challenges

6

Customer retention challenges:

With many alternatives available, keeping customers loyal is increasingly difficult.

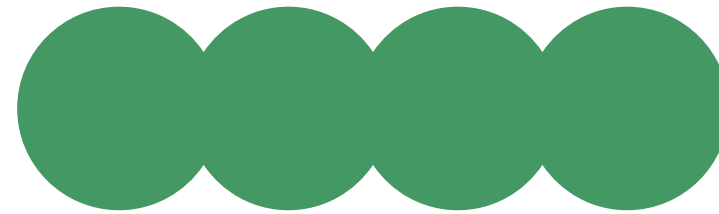
7

Post-Purchase Experience:

Poor delivery, returns, or service reduce satisfaction and future conversion likelihood.

Other Challenges Facing the Digital Retail Industry





Main Problem: **Low Purchase Conversion in Digital Retail**

Digital retail platforms collect large amounts of customer behavior data.

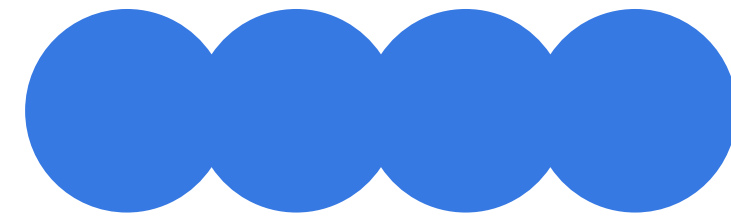
However, many businesses still struggle to understand why customers browse products but leave without buying.

This creates low purchase conversion, where customer interest does not turn into completed sales.

As a result, businesses lose potential revenue and miss opportunities to improve customer experience.



**Key issue: customer hesitation and
cart abandonment.**



Main goal+Objective

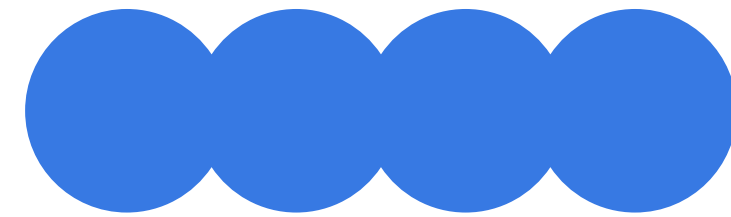


Analyze customer behavior to understand why customers browse or add items to cart but do not complete purchases.

Main Goal

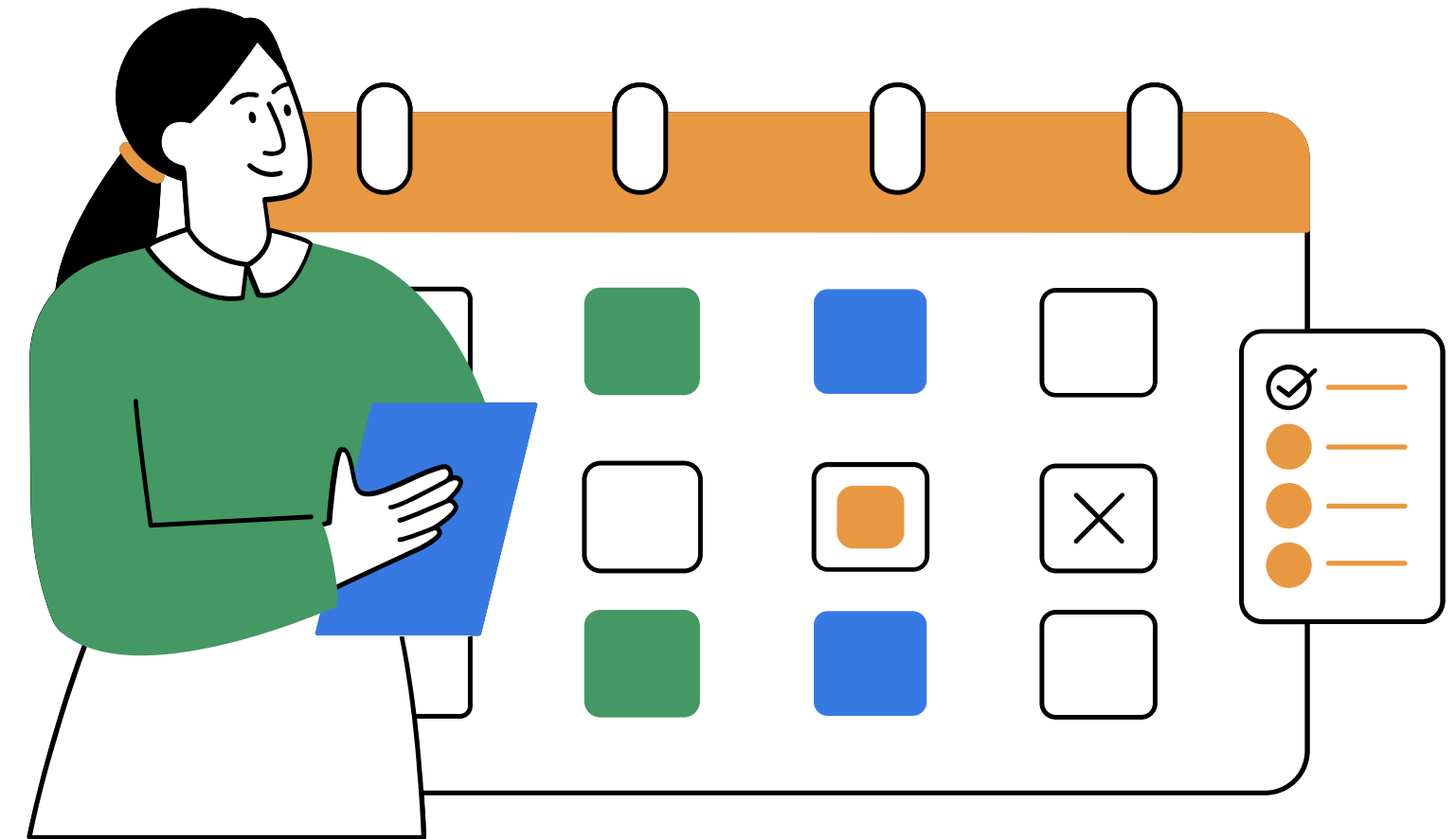
Objectives

- Identify patterns in **product views, cart activity, and purchase behavior**.
- Compare **cart abandonment trends** across product categories.
- Segment customers based on **shopping behavior and purchase intent**.
- Recommend strategies to **improve conversion rates** and reduce lost sales.



Data Collection

Data Sources



Kaggle

- Platform for datasets and data science competitions
- Mix of real-world and user-uploaded datasets
- Good for practical and exploratory projects

Huggingface

- Repository for AI models and large-scale datasets
- Often used in research and machine learning
- Includes text, behavioral, and product data

UC Irvine Machine Learning Repository

- Academic database of curated datasets
- Typically clean, structured, and well-documented
- Commonly used for benchmarking and research



McAuley-Lab
/Amazon-Reviews-2023

huggingface.co

McAuley-Lab/Amazon-Reviews-2023 • Datasets at Hugging Face

We're on a journey to advance and democratize artificial intelligence through open source and open science.

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UCI Machine Learning Repository

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Datasets

Official Real Data



UCI Machine Learning Repository

Discover datasets around the world!

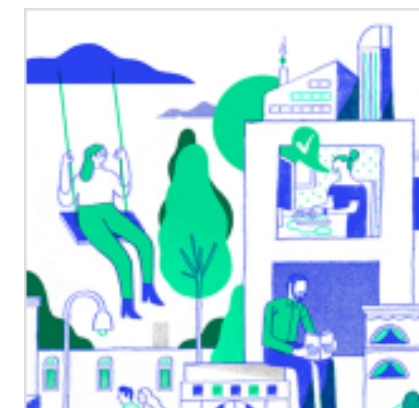
ics.uci.edu



Instacart Market Basket Analysis

Discover what actually works in AI. Join millions of builders, researchers, and labs evaluating agents, models, and frontier technology through crowdsourced benchmarks,...

kaggle.com



Brazilian E-Commerce Public Dataset by Olist

100,000 Orders with product, customer and reviews info

kaggle.com



Ecommerce Clickstream Dataset

Raw clickstream logs for analysis and modeling

[k](#) kaggle.com



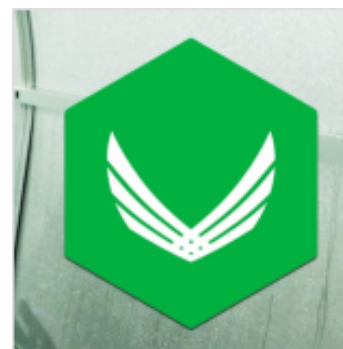
Online Shoppers Purchasing Dataset

Online shoppers session dataset with 1M records for purchase prediction

[k](#) kaggle.com

Datasets

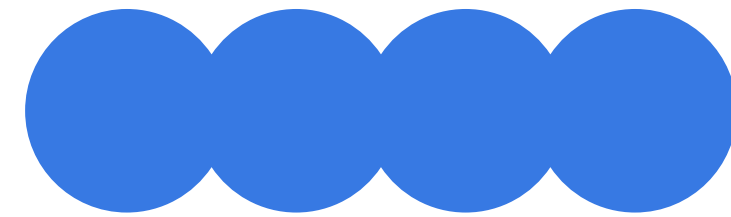
User-uploaded data



eCommerce behavior data from multi category store

This dataset contains 285 million users' events from eCommerce website

[k](#) kaggle.com



Input, Process, and Output

Input

The input for this project will include:



Customer behavior data

Product views, page visits, cart additions, and purchase events



Product Information

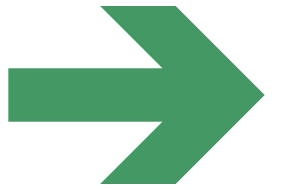
Product attributes such as price, category, and availability that influence purchase decisions.



Customer Session Data

User activity, timing, sequence of actions, and conversion flow.

Process



Key points:

➔ *Data Preparation*

- Clean and prepare the customer behavior dataset
- Remove missing, duplicate, or inconsistent records

➔ *Cart Abandonment Analysis*

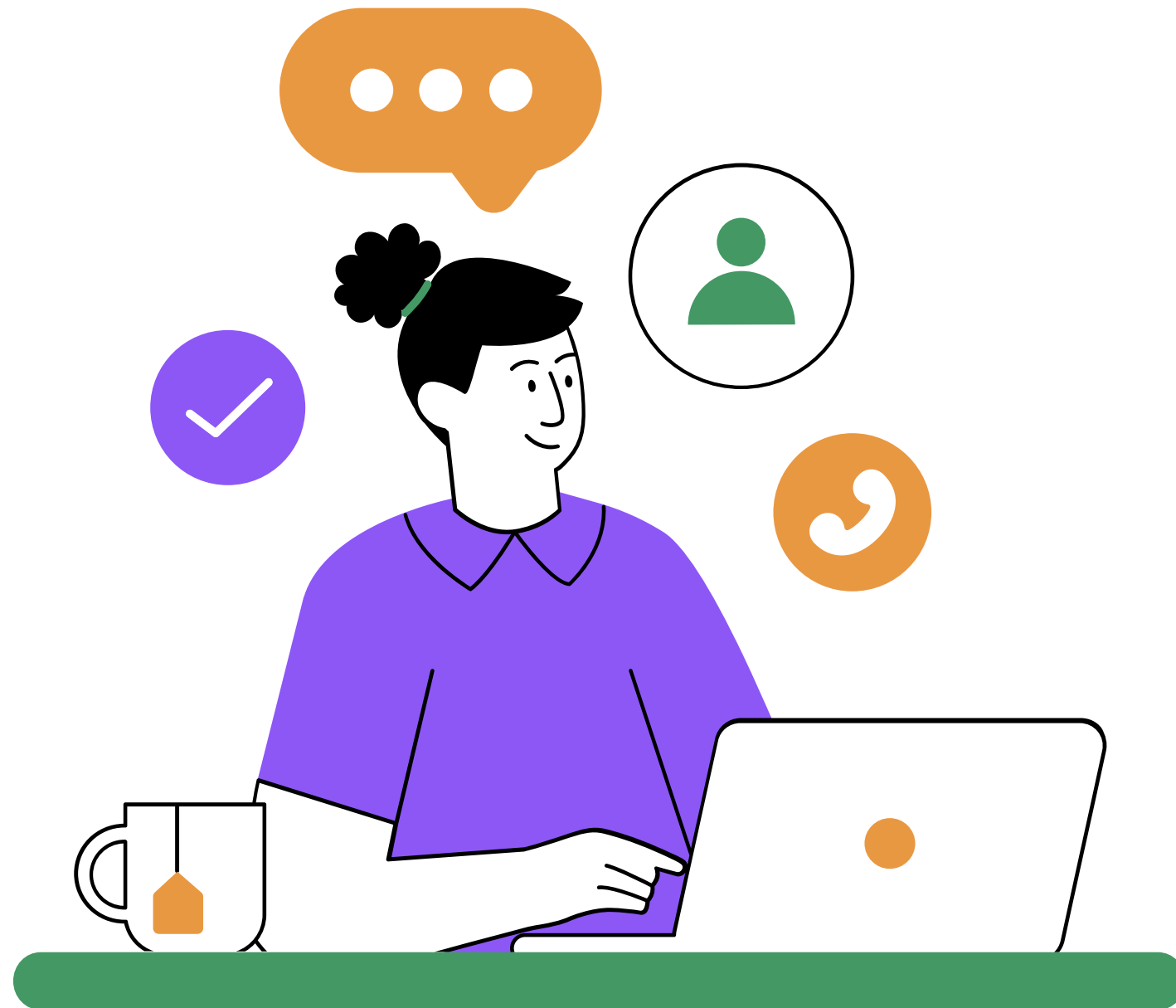
- Analyze abandonment patterns across different product categories
- Compare which categories have higher hesitation or drop-off rates

➔ *Customer Journey Mapping*

- Track how customers move from browsing → cart → purchase or abandonment
- Identify where customers drop off in the buying process



Process



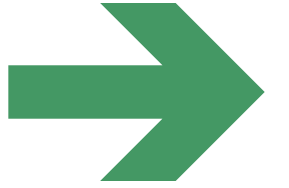
➔ ***Customer Segmentation***

Group customers by behavior, such as:

- High-interest browsers
- Price-sensitive customers
- Likely abandoners
- High-intent buyers

➔ ***Predictive Modeling***

- Build a model to predict which customers are likely to abandon their cart
- Use the results to support targeted business strategies, such as discounts or reminder emails.

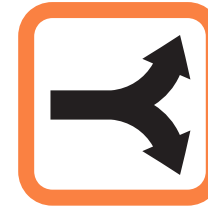


Output



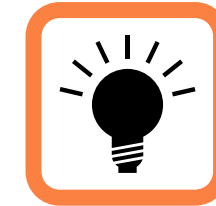
Customer Behavior Insights

- Understand how customers interact with products across different categories
- Identify patterns in browsing, cart activity, and purchase decisions



Cart Abandonment Findings

- Identify product categories with higher cart abandonment rates
- Understand where customers are dropping off before checkout



Customer Segments

Group customers into meaningful segments, such as:

- High-hesitation customers
- Price-sensitive customers
- Frequent browsers
- High-intent buyers

Output



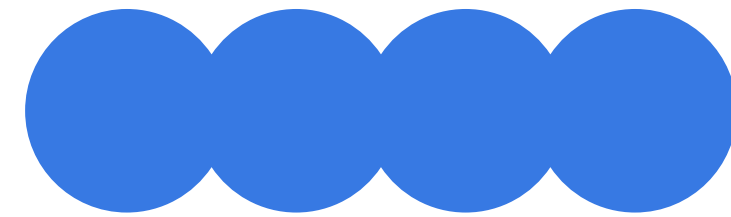
Business Recommendations:

- Suggest targeted strategies such as:
- Discount offers
- Reminder emails
- Free shipping options
- Better product reviews or product information
- Personalized product recommendations



Retail Strategy Support:

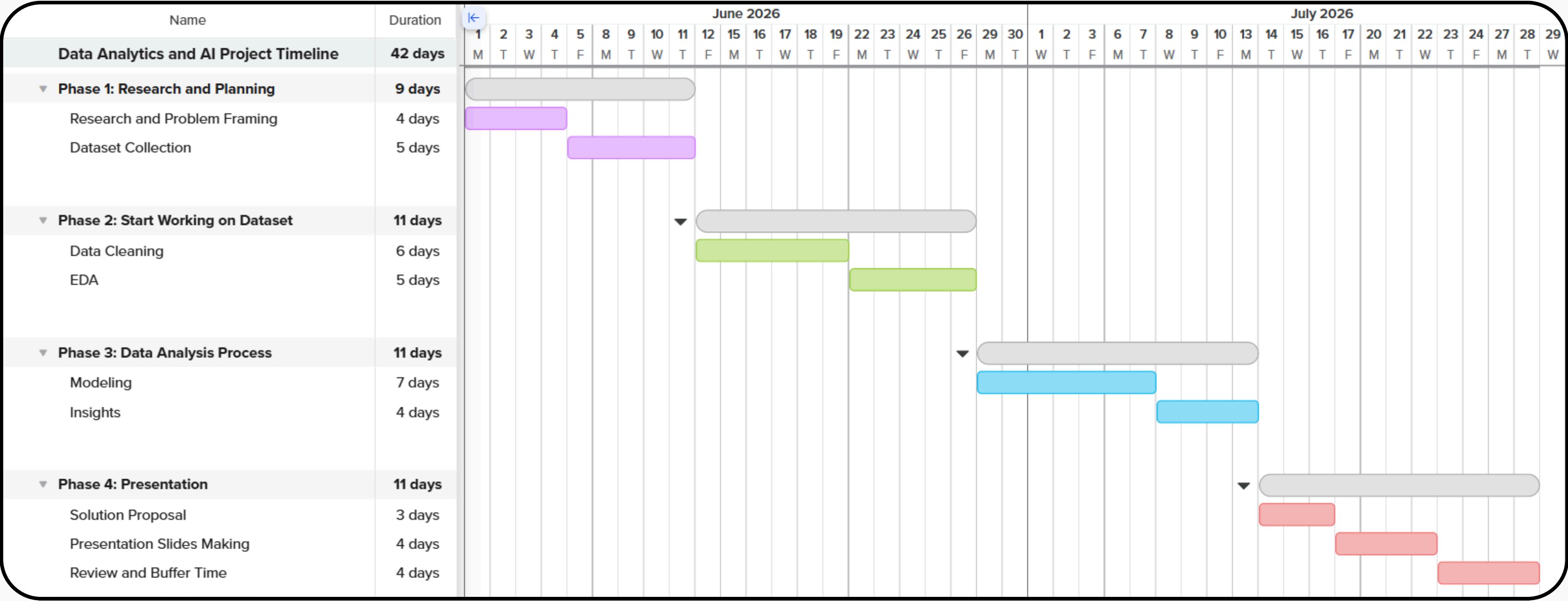
- Help businesses decide which products to focus on
- Support inventory planning by identifying products that may be overstocked or underperforming
- Improve conversion rates and reduce lost sales.



Timeline and Pricing

Data Analysis Project Timeline

(6 weeks total)



Section 6 of 6

McKinsey & Company

GSA Schedule

GSA Rates for Services

SINs Awarded	Labor Category/Service Title	Unit of Issue	GSA Price (including IFF)
541611, 541990	Senior Partner - Executive/ Strategy	Hour	\$1,193.57
541612HC and 54151S			
541611, 541990	Partner- Executive/ Strategy	Hour	\$1,072.02
541612HC and 54151S			
541611, 541990	Associate Partner- Executive/ Strategy	Hour	\$958.13
541612HC and 54151S			
541611, 541990	Engagement Manager- Executive/ Strategy	Hour	\$834.40
541612HC and 54151S			
541611, 541990	Associate- Executive/ Strategy	Hour	\$498.23
541612HC and 54151S			
541611, 541990	Business Analyst- Executive/ Strategy	Hour	\$339.45
541612HC and 54151S			
541611, 541990 and 54151S	Associate - Data Science	Hour	\$480.71
541611, 541990 and 54151S			
541611, 541990 and 54151S	Business Analyst - Data Science	Hour	\$327.41
541611, 541990 and 54151S			
541611, 541990	Associate - Human-Centered Design	Hour	\$480.71
541611, 541990			
541611, 541990	Business Analyst - Human-Centered Design	Hour	\$327.41
541611, 541990			
541611, 541990	Associate - Implementation	Hour	\$465.38
541612HC and 54151S			
541611, 541990	Business Analyst - Implementation	Hour	\$317.55
541612HC and 54151S			
541611, 541990	Media Designer	Hour	\$182.86
541612HC and 54151S			
541611, 541990	Graphics Project Manager	Hour	\$171.92
541612HC and 54151S			
541611, 541990	Business Presentation Designer	Hour	\$162.06
541612HC and 54151S			
541611, 541990	Capabilities & Insights Expert	Hour	\$366.83
541612HC and 54151S			
541611, 541990	Capabilities & Insights Analyst	Hour	\$275.94
541612HC and 54151S			
541611, 541990	Engagement Team Assistant	Hour	\$120.45
541612HC and 54151S			

GSA APPROVED PRICELIST
THE BOSTON CONSULTING GROUP, INC.

The rates below include the Industrial Funding Fee (IFF)

SIN	Awarded Labor Category	Year 3	Year 4	Year 5
541611, 541690, 541715, 541990	Executive Strategy Senior Partner	\$1,116.17	\$1,192.07	\$1,273.13
541611, 541690, 541715, 541990	Executive Strategy Director	\$1,050.88	\$1,122.34	\$1,198.66
541611, 541690, 541715, 541990	Executive Strategy Project Leader	\$711.35	\$759.72	\$811.38
541611, 541690, 541715, 541990	Executive Strategy Senior Consultant	\$629.35	\$672.14	\$717.84
541611, 541690, 541715, 541990	Executive Strategy Consultant	\$479.91	\$512.54	\$547.40
541611, 541690, 541715, 541990	Executive Strategy Senior Associate	\$404.18	\$431.67	\$461.02
541611, 541690, 541715, 541990	Lead Knowledge Analyst	\$321.59	\$343.46	\$366.81
541611, 541690, 541715, 541990	Knowledge Analyst	\$178.82	\$190.98	\$203.97
541611, 541690, 541715, 541990	Senior Experience Designer	\$324.11	\$346.15	\$369.68
541611, 541690, 541715, 541990	Senior Engineer	\$204.97	\$218.90	\$233.78
541611, 541690, 541715, 541990	Principal Data Scientist	\$310.98	\$332.13	\$354.72
541611, 541690, 541715, 541990	Lead Data Scientist	\$202.14	\$215.88	\$230.56
541611, 541690, 541715, 541990	Data Scientist	\$129.34	\$138.14	\$147.53

The Boston Consulting Group, Inc.
(301) 664-7400

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KEARNEY

GSA Schedule

GSA Pricing

The rates below include the Industrial Funding Fee (IFF).

SIN	Labor Category	GSA Price
541611	Executive Strategy Project Leader	\$624.00
	Executive Strategy Senior Consultant	\$540.02
	Strategy Consultant	\$421.00
	Junior Consultant	\$233.20
	Functional Expert and Researcher	\$118.00
	Production and Design Expert	\$96.00
	Research Analyst	\$90.10
	Graphic Designer	\$76.32
	Project Support Admin	\$51.00
	Executive Consultant Sr.	\$671.29
	Project Director Sr.	\$572.09
	Program Manager Sr.	\$453.59
	Program Manager Jr.	\$401.18
	Management Consultant Sr.	\$343.88
	Management Consultant Jr.	\$294.89
	Analyst Sr.	\$225.56

Kearney PSDS acknowledges that the Service Contract Labor Standards (SCLS) is applicable to this contract as it applies to the entire Multiple Award Schedule and all services provided. While no specific labor categories have been identified as being subject to SCLS due to exemptions for professional employees (FAR 22.1101, 22.1102 and 29 CFR 541.300), this contract still maintains the provisions and protections for SCLS eligible labor categories. If and / or when Kearney adds SCLS labor categories / employees to the contract through the modification process, Kearney must inform the Contracting Officer and establish a SCLS matrix identifying the GSA labor category titles, the occupational code, SCLS labor category titles and the applicable WD number. Failure to do so may result in cancellation of the contract.

Consulting Rate Benchmark

- Big consulting firms can charge very high hourly rates
- We used GSA-approved consulting rates as a pricing benchmark
- GSA rates are public hourly prices used for U.S. government consulting contracts.
- We compared rates from major consulting firms:
- McKinsey & Company
- Boston Consulting Group, Inc.
- Kearney

McKinsey

- Engagement Manager: \$834.40/hr
- Associate, Data Science: \$480.71/hr
- Business Analyst, Data Science: \$327.41/hr

BCG

- Principal Data Scientist: \$310.98/hr
- Data Scientist: \$129.34/hr

Kearney

- Executive Strategy Project Leader: \$624.00/hr
- Strategy Consultant: \$421.00/hr
- Management Consultant Jr.: \$294.89/hr
- Analyst Sr.: \$225.56/hr

Pricing Comparison



Man Hour Cost Assumption:

Roles	Rate
Project Manager / Lead Consultant	¥23,400/hr
Data Scientist	¥21,840/hr
Business Consultant	¥15,600/hr
Data Analyst	¥20,280/hr
Reporting Support	¥12,480/hr

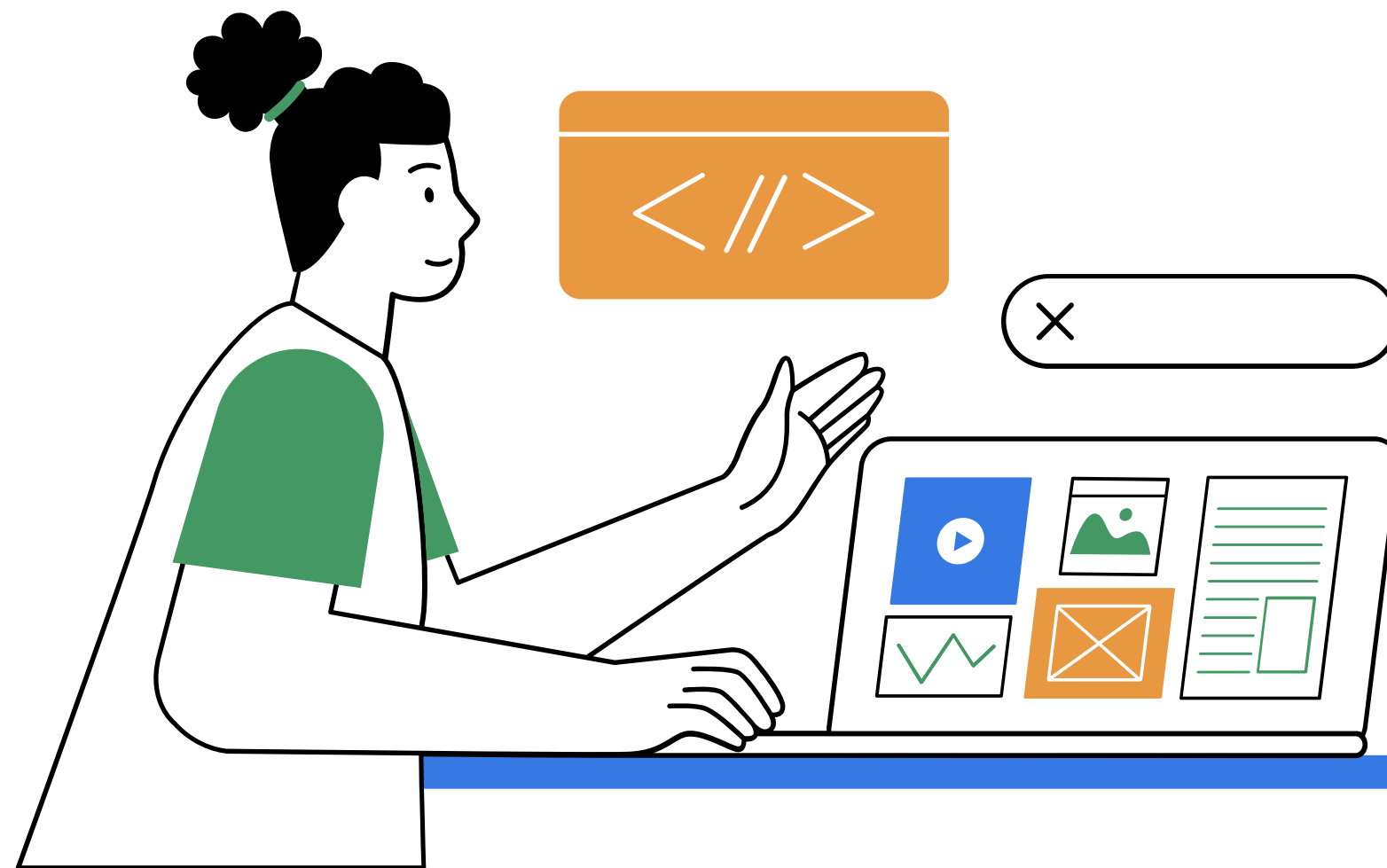
Cost

Total Quotation after 35% Consulting Markup = **¥4,948,944× 135%**
= **¥6,681,074**

Detailed Cost:

Phase	Roles Required	Hours	Estimated Cost
Phase 1: Research and Planning	Project Manager, Business Consultant, Data Analyst	54hrs	¥1,042,080
Phase 2: Dataset Work	Data Analyst, Data Scientist	66hrs	¥1,110,720
Phase 3: Data Analysis Process	Data Scientist, Business Consultant, Project Manager	72hrs	¥1,478,880
Phase 4: Presentation	Business Consultant, Reporting Specialist, Project Manager	52hrs	¥867,360
Total		244hrs	¥4,499,040
Contingency, 10%	Extra time for revisions/unexpected issues		¥449,904
Final Amount			¥4,948,944

Reference



- <https://slideworks.io/resources/management-consulting-fees-how-mc-kinsey-prices-projects>
- https://colab.research.google.com/drive/10ljqNJrED-c_wOReUPRIWAFQLHyw-o3?usp=sharing